Calculus I - Derivatives

The derivative measures instantaneous rate of change.

Power rule: d/dx x^n = n x^(n-1)

Chain rule: d/dx f(g(x)) = f'(g(x)) g'(x)

Example: f(x) = (3x^2 + 1)^4

Exercise: use the chain rule to find the derivative.

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Worked solution

Let u = 3x^2 + 1. Then du/dx = 6x.

The derivative of u^4 is 4u^3.

Multiply: f'(x) = 24x(3x^2 + 1)^3.

Check: the outside power drops by one.